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[MANDATORY NOTICE]

SYNTHETIC DEMONSTRATION DATA

This document is fictional test data created for the VYASA prototype.

It is not an actual operating procedure, engineering standard,
regulatory requirement, or safety instruction.

All numerical values are example synthetic demonstration values only.

SECTION 1: SCOPE, TERMINOLOGY & CONCEPTS

1.1 Scope & Purpose

This document establishes synthetic terminology and operational concepts for elevated refinery flare headers, knockout drums, and water seal drums in a fictional process plant.

1.2 Flare Header Terminology

- Flare Header: Main collecting pipe conveying emergency relief gases to the flare stack.
- Knockout Drum (KOD): Dedicated separation vessel removing entrained liquid droplets prior to combustion.
- Water Seal Drum: Liquid barrier preventing flame flashback from the stack into upstream headers.
- Flare Tip Seal: Purge reduction seal positioned at the stack discharge.
- Continuous Purge Gas: Continuous supply of non-condensable gas maintaining positive outflow.

1.3 Purge Gas Concept

Positive purge gas flow is continuously required through the flare header to maintain positive internal pressure, counteract wind-induced atmospheric air ingress, and eliminate the possibility of forming an explosive hydrocarbon-oxygen mixture inside the flare stack riser.

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SECTION 2: PURGE VELOCITY OPERATING LIMITS (SYNTHETIC DEMONSTRATION)

2.1 Purge Gas Velocity Concepts & Example Operational Limits

Purge gas flow rates must be measured continuously at the stack base and seal discharge.

Example / Synthetic Demonstration Values:

- Example synthetic demonstration minimum continuous purge velocity for fuel gas header: 0.04 m/s.
- Example synthetic demonstration minimum continuous purge velocity for hydrogen flare seal: 0.03 m/s.
- Example synthetic demonstration recommended continuous operating velocity range: 0.03 m/s to 0.15 m/s.
- Example synthetic demonstration maximum design velocity limit during emergency relief depressurization: 12.0 m/s.

2.2 Regulatory & Legal Notice Regarding Limits

IMPORTANT: The figures stated above are example synthetic demonstration values created exclusively for software agent testing and verification within the VYASA prototype. They do NOT represent statutory regulatory limits, official API 521 standards, or MRPL operating specifications. No real-world regulatory compliance or operational validity may be inferred from this fictional demonstration document.

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SECTION 3: ALARM THRESHOLDS & FLARE HEADER OBSERVATIONS

3.1 Example Alarm & Monitoring Thresholds

- Example synthetic demonstration low purge flow alarm: triggered when purge velocity drops below 0.02 m/s.
- Example synthetic demonstration pilot flame loss warning: thermocouple reading below 200.0 °C.
- Example synthetic demonstration KOD high-high liquid level trip: > 65% liquid volume in knockout drum.

3.2 Flare Header Observations & Field Safety Practices

- Smokeless flaring steam ratio observations: steam injection flow should maintain clear plume without over-steaming.
- Header pressure observation: maintain positive pressure above 5.0 mbarg to prevent negative draft.
- Visual inspection: observe flare tip flame stability and absence of internal burning signs.